

Equivariant cohomology

1. Poincaré duality

Let M be compact and oriented, and let $[M]$ be its fundamental class. Then

$$\begin{aligned} H^p(M) &\xrightarrow{\sim} H_{n-p}(M), \\ \alpha &\mapsto [M] \cap \alpha \end{aligned}$$

is an isomorphism.

If M is noncompact, then

$$H_c^p(M) \simeq H_{n-p}(M).$$

- $H_c^p(M) = \varinjlim_{\substack{K \subseteq M \\ K \text{ compact}}} H^p(M, M \setminus K).$
- In terms of de Rham cohomology: differential forms with compact support.

Definition. Let $f: M \rightarrow N$, where M, N are compact and oriented. The *Gysin map* (cohomology pushforward)

$$f_!: H^p(M) \longrightarrow H^{p+n-m}(N), \quad n - m = \dim N - \dim M \approx \text{codim } M (?),$$

is given by

$$f_!(\alpha) = \text{PD}_N \circ f_*^{\text{hom}} \circ \text{PD}_M(\alpha).$$

There are two important cases.

(1) Fiber bundle case = fiber integral. If $\pi: E \rightarrow B$ is a fiber bundle with fiber dimension r , then

$$\pi_!: H^p(E) \longrightarrow H^{p-r}(B).$$

In de Rham cohomology,

$$\pi_!([\beta]) = \left[\int_{E/B} \beta \right],$$

where $\int_{E/B} \beta$ is the fiber integral of a p -form.

(2) Embedding case $\approx$ Thom class. If $i: Y \hookrightarrow M$, let N be the normal bundle. Then a tubular neighborhood of Y is diffeomorphic to N .

In general, if $E \rightarrow Y$ is a vector bundle of rank k , define the *Thom class* $[\Phi] \in H_c^k(E)$ by

$$[\Phi] = \text{PD}_E(i(Y)),$$

where $i: Y \hookrightarrow E$ is given by the zero section.

In de Rham cohomology, let Φ be a closed form on E representing the Thom class (i.e., a *Thom form*). Then

$$\int_{E_y} \Phi = 1 \quad \text{for all } y \in Y.$$

Thom isomorphism.

$$\boxed{H^k(Y) \xrightarrow{\sim} H_c^{k+r}(E), \quad \alpha \mapsto \pi^* \alpha \cup [\Phi].}$$

Coming back to $i: Y \hookrightarrow M$, with

$$N = \frac{TM}{TY} \longrightarrow Y,$$

consider the Thom class on N , denoted by Φ_Y . Then

$$i_!: H^p(Y) \longrightarrow H^{p+k}(M)$$

is given by (exercise)

$$i_!(\alpha) \mapsto \pi_N^* \alpha \cup [\Phi_Y],$$

i.e.,

$$i_!: H^k(Y) \xrightarrow[\text{iso}]{\text{Thom}} H^{k+r}(N, N \setminus Y) \longrightarrow H^{k+r}(M).$$

In particular,

$$[\Phi_Y] = i_!(1) = \text{PD}_M \circ i_*([Y]) = \text{PD}_M([Y]).$$

(Extend Φ by zero from a tubular neighborhood of Y to M .)

$\therefore$ Thom class of the normal bundle = Poincaré dual.

Finally, given $E \rightarrow Y$, $[\Phi]$, and the zero section $i: Y \hookrightarrow E$, define the *Euler class* $e(E) \in H^r(Y)$ by

$$e(E) := i^*[\Phi].$$

If E is a complex vector bundle of complex rank s , then

$$e(E) = c_s(E) = \text{PD}(\text{zero locus of a generic (transverse to zero section) section of } E).$$

추가 Topic

If $Y \hookrightarrow X$ is a complex hypersurface, denote the holomorphic normal bundle by

$$N'_Y := \frac{T'X|_Y}{T'Y}.$$

This is a line bundle.

Adjunction formula.

$$N'_Y \simeq \mathcal{O}(Y)|_Y,$$

where Y is considered as a divisor. This implies

$$K_Y \simeq (K_X \otimes \mathcal{O}(Y))|_Y.$$

2. Equivariant cohomology

(Why? Don't ask me...)

$$H^*: \mathbf{Top} \longrightarrow \mathbf{Rng} \implies H_G^*: G\text{-}\mathbf{Top} \longrightarrow \mathbf{Rng},$$

such that, if $G \curvearrowright M$ and M/G is good enough, with no stabilizer (= free), then

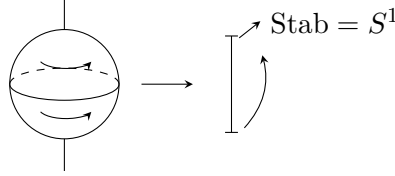
$$H_G^*(M) = H^*(M/G).$$

Examples.

- (1) $\mathbb{Z} \curvearrowright \mathbb{R}$, $x \mapsto x + n$. Then

$$\mathbb{R}/\mathbb{Z} \simeq S^1, \quad H_{\mathbb{Z}}^*(\mathbb{R}) = H^*(S^1). \quad \text{Good!}$$

- (2) $S^1 \curvearrowright S^2$ by rotation.



$$H^*(S^2/S^1) = H^*(I) = H^*(\text{pt}): \quad \text{bad}\cdots$$

Therefore, when the action is not free, we need to record the information of the stabilizer.

Lemma. If $G \curvearrowright E$ freely and $G \curvearrowright M$, then the diagonal action $G \curvearrowright E \times M$ is free.

If EG is contractible and $G \curvearrowright EG$ freely, then $EG \times M$:

- (1) has the same homotopy type as M ;
- (2) admits a free G -action.

Fact. For every compact Lie group G , there exists a principal G -bundle

$$EG \longrightarrow BG$$

with EG contractible. In particular, $EG \curvearrowright G$ freely. It is called a *universal G -bundle*. The space BG is called a *classifying space for G* . That is:

- (1) Let X be a CW complex and $P \rightarrow X$ a principal G -bundle. Then there exists $h: X \rightarrow BG$ such that

$$P \simeq h^*EG.$$

- (2) If $h_0, h_1: X \rightarrow BG$ and $h_0^*EG \simeq h_1^*EG$, then $h_0 \simeq h_1$ are homotopic.

That is,

$$[X, BG] \simeq P_G(X).$$

Definition. The *homotopy quotient* of M by G is

$$M_G := EG \times M/G,$$

where

$$(e, x) \sim (e \cdot g^{-1}, gx), \quad g \in G.$$

Definition. The *equivariant cohomology* of M by G is

$$H_G^*(M; R) := H^*(M_G; R), \quad R: \text{coefficient ring.}$$

- (1) It is well-defined.
- (2) If $G \curvearrowright M$ freely, then $M_G \simeq_w M/G$.

Proof. Cartan's mixing diagram:

$$\begin{array}{ccccc} P & \longleftarrow & P \times M & \longrightarrow & \boxed{M} \\ \text{prin.} \downarrow & & \downarrow \text{prin.} & & \downarrow \\ \boxed{B} & \longleftarrow & \boxed{P \times_G M} & \longrightarrow & M/G \end{array}$$

If $P \rightarrow B$ is principal, then $P \times_G M \rightarrow B$ is a fiber bundle with fiber M .
For (2), we use the following fact.

Fact. If $G \curvearrowright M$ freely and M is a manifold, then

$$M \longrightarrow M/G$$

is a principal G -bundle.

By symmetry,

$$\begin{array}{ccccc} \boxed{P} & \longleftarrow & P \times M & \longrightarrow & M \\ \downarrow & & \downarrow & & \downarrow \text{prin.} \\ B & \longleftarrow & \boxed{P \times_G M} & \longrightarrow & \boxed{M/G} \end{array}$$

If P is contractible, then

$$P \times_G M \simeq_{(w)} M/G \quad (\text{weakly homotopic})$$

by the homotopy exact sequence. □

(1)도 비슷하게 됨.

Example. $G \curvearrowright \text{pt}$ trivially. Then

$$\text{pt}_G = EG \times \text{pt}/G = BG,$$

so

$$H_G^*(\text{pt}) = H^*(BG).$$

Consider the constant map $M \rightarrow \text{pt}$, which is a G -map. Then

$$\pi_G: M_G \longrightarrow \text{pt}_G = BG,$$

and hence

$$\pi_G^*: H^*(BG) \longrightarrow H_G^*(M).$$

Therefore $H_G^*(M)$ is an $H^*(BG)$ -algebra (module).

(i.e., $H^*(BG) \simeq \text{coefficient ring}$).

Example: BS^1 and ES^1 . The action

$$S^1 \curvearrowright S^{2n-1} \subset \mathbb{R}^{2n} = \mathbb{C}^n$$

is free. Hence

$$S^{2n-1} \longrightarrow S^{2n-1}/S^1 \simeq \mathbb{C}P^{n-1}$$

is a principal S^1 -bundle.

Consider

$$\mathbb{C}^1 \subset \mathbb{C}^2 \subset \mathbb{C}^3 \subset \dots$$

This gives the commutative diagram

$$\begin{array}{ccccccc} S^1 & \hookrightarrow & S^3 & \hookrightarrow & S^5 & \hookrightarrow & \dots \\ \downarrow & & \downarrow & & \downarrow & & \\ \mathbb{C}P^0 & \longrightarrow & \mathbb{C}P^1 & \longrightarrow & \mathbb{C}P^2 & \longrightarrow & \dots \end{array}$$

Let

$$S^\infty = \bigcup S^{2n+1}, \quad \mathbb{C}P^\infty = \bigcup \mathbb{C}P^n.$$

Then S^1 acts freely on S^∞ , and

$$S^\infty \longrightarrow \mathbb{C}P^\infty$$

is a principal S^1 -bundle. Since S^∞ is contractible, $S^\infty \rightarrow \mathbb{C}P^\infty$ is a universal S^1 -bundle. In particular,

$$\begin{aligned} BS^1 &= \mathbb{C}P^\infty, & H^*(BS^1) &= H^*(\mathbb{C}P^\infty) = \mathbb{Z}[u]. \\ & & (H^*(\mathbb{C}P^n) &= \mathbb{Z}[u]/(u^{n+1})). \end{aligned}$$

For $\mathbb{T} := (\mathbb{C}^*)^{m+1}$,

$$\mathbb{C}^\infty - \{0\} \longrightarrow \mathbb{C}P^\infty = B\mathbb{C}^*,$$

and

$$H_{\mathbb{T}}^*(\text{pt}) = H^*((\mathbb{C}P^\infty)^{m+1}) = \mathbb{Z}[t_0, \dots, t_m].$$

3. Cartan model (= de Rham model)

Let $\mathfrak{g} := \text{Lie}(G)$. Define

$$\Omega_G(M) := [S(\mathfrak{g}^\vee) \otimes \Omega(M)]^G = \{\text{equivariant differential forms}\}.$$

The G -action on $S(\mathfrak{g}^\vee) \otimes \Omega(M)$ is given on $\Omega(M)$ by the action induced by $G \curvearrowright M$ ($\mathcal{L}$), and on $S(\mathfrak{g}^\vee)$ by the adjoint action.

We can consider $S(\mathfrak{g}^\vee) \otimes \Omega(M)$ as $\Omega(M)$ -valued polynomial maps on $\mathfrak{g}$. Then $\Omega_G(M)$ consists of the G -equivariant such maps. Grading: $\deg u = 2$ for $u \in \mathfrak{g}^\vee$ and degree of differential forms.

Cartan differential. Define $D : \Omega_G(M) \rightarrow \Omega_G(M)$ by

$$(D\alpha)(X) = d(\alpha(X)) - \iota_X(\alpha(X)), \quad X \in \mathfrak{g}.$$

Theorem (equivariant de Rham theorem).

$$H_G^*(M; \mathbb{R}) \simeq H^*(\Omega_G(M), D).$$

For $G = T = (S^1)^\ell$, the adjoint action is trivial. Hence

$$\begin{aligned}\Omega_T(M) &= S(\mathfrak{t}^\vee) \otimes \Omega(M)^T \\ &= \Omega(M)^T[u_1, \dots, u_\ell],\end{aligned}$$

where the u_i are generators of $\mathfrak{t}^\vee$. If $\alpha \in \Omega_T(M)$, then

$$\alpha = \sum_I u^I \alpha_I, \quad \alpha_I \in \Omega(M)^T, \quad u^I = u_1^{i_1} \cdots u_\ell^{i_\ell}.$$

Consequently,

$$D\alpha = d\alpha - \sum_i u_i \iota_i \alpha, \quad (du_i = 0, \quad \iota_i u_j = 0).$$

Here $\iota_i = \iota_{X_i}$, and $\{X_i\}$ and $\{u_i\}$ are dual.

If $D\alpha = 0$, then α is *equivariantly closed*.

Example. Let $S^1 \curvearrowright M$, and let ϕ be an equivariant form of degree $2n$. Then

$$\phi = \phi_{2n} + \phi_{2n-2}u + \cdots + \phi_2 u^{n-1} + \phi_0 u^n, \quad \deg \phi_k = k.$$

We have

$$D\phi = d_X \phi = 0 \iff (d - u\iota_X)\phi = 0 \iff \begin{cases} d\phi_{2n} = 0, \\ \iota_X \phi_{2n} = d\phi_{2n-2}, \\ \vdots \\ \iota_X \phi_2 = d\phi_0. \end{cases}$$

In particular, if $n = 1$, then

$$\phi = \omega + u \cdot H$$

is equivariantly closed if and only if

$$d\omega = 0, \quad \iota_X \omega = dH,$$

i.e., $S^1 \curvearrowright (M, \omega)$ is Hamiltonian with moment map H .

Example. Another important example of equivariantly closed classes:

Let $\pi: E \rightarrow M$ be a G -equivariant vector bundle. Then

$$\pi_G: E_G \longrightarrow M_G$$

is a vector bundle of the same rank.

Define *equivariant characteristic classes* by

$$c_i^G(E) := c_i(E_G) \in H^*(M_G) = H_G^*(M).$$

In particular, the *equivariant Euler class* is

$$e^G(E) := e(E_G).$$

The class $c_i^G(E)$ is an equivariantly closed extension of $c_i(E)$.

Example. If $S^1 \curvearrowright (M, \omega)$, does there exist a moment map? That is, does there exist an equivariantly closed extension

$$\omega \longmapsto \omega + uH \quad \text{such that} \quad d_X(\omega + uH) = 0?$$

Yes, if $[\omega] = c_1(L)$ and $L \rightarrow M$ is S^1 -equivariant. Then

$$c_1^{S^1}(L) = [\omega + uH]. \quad \square$$

4. Localization

1. Borel. Let

$$T = (S^1)^r, \quad R = H^*(BT, \mathbb{Q}) = \mathbb{Q}[u_1, \dots, u_r].$$

Consider $\chi \in \mathbb{Z}^r \setminus \{0\}$, $\chi = (m_1, \dots, m_r)$, and

$$\lambda_\chi = m_1 u_1 + \dots + m_r u_r$$

(a nonzero homogeneous linear polynomial). Then

$$S = \{\lambda_{\chi_1} \cdots \lambda_{\chi_k}; \ k \geq 0, \ \chi_j \in \mathbb{Z}^r \setminus \{0\}\}$$

is a multiplicative subset of R .

Recall that $H_T^*(M)$ is an R -module.

Fact. If M is an R -module and $S \subset R$ is multiplicative, then

$$M_S \simeq M \otimes_R R_S, \quad \frac{ax}{s} \longleftrightarrow x \otimes \frac{a}{s}.$$

Theorem (Borel localization). Let $F = M^T$, i.e., the fixed points of $T \curvearrowright M$, and $i: F \hookrightarrow M$. Then

$$i_S^*: H_T^*(M)_S \xrightarrow{\sim} H_T^*(F)_S$$

is an isomorphism.

Recall that, for an R -module M ,

$$\ker(M \longrightarrow M_S)$$

is the S -torsion. Indeed, if $s \in S$ and $s \cdot x = 0$, then

$$x = s^{-1} \cdot s \cdot x = 0 \quad \text{in } M_S.$$

Thus, equivalently,

$$H_T^*(M) \xrightarrow{i^*} H_T^*(F)$$

is an isomorphism up to torsion.

Remark.

$$S \subset \mathbb{Q}(u_1, \dots, u_r) = \text{Frac}(\mathbb{Q}[u_1, \dots, u_r]).$$

2. ABBV. Consider $i: F \hookrightarrow M$ (not necessarily $F = M^T$). Recall

$$i_!: H^k(F) \xrightarrow{\text{Thom}} H^{k+r}(M, M \setminus F) \longrightarrow H^{k+r}(M),$$

$$i_![\beta] = [\pi^* \beta \wedge \Phi_N].$$

Under the identification of N_F with a tubular neighborhood of F , i is the zero section s . Hence

$$i^* i_!(\beta) = s^*(\pi^* \beta \cup \Phi_N) = \beta \cup e(N).$$

Therefore the self-intersection formula is

$$i^* i_*(\beta) = e(N) \cup \beta.$$

The same holds for equivariant cohomology. If $T \curvearrowright M$, $F = M^T$, then normal bundle N is a T -equivariant vector bundle, hence

$$N_T \longrightarrow F_T = (ET \times F)/T,$$

and

$$\boxed{i^* i_*(\beta) = e_T(N) \cup \beta}, \quad \beta \in H_T^*(F).$$

(Just put “equivariant” everywhere.)

Borel localization gives

$$H_T^*(M)_S \xrightarrow[\sim]{i^*} H_T^*(F)_S.$$

By [AB], $e_T(N)$ is invertible in $H_T^*(F)_S$.

Claim. For every $\phi \in H_T^*(M)$,

$$\boxed{\phi = \sum_F i_* \left(\frac{i^* \phi}{e_T(N)} \right)}. \quad (4.3)$$

Proof. Let

$$\gamma := i_*(e_T(N)^{-1} \cup i^* \phi) \in H_T^*(M)_S.$$

Then

$$\begin{aligned} i^* \gamma &= i^* i_*(e_T(N)^{-1} \cup i^* \phi) \\ &= e_T(N) \cup e_T^{-1}(N) \cup i^* \phi = i^* \phi. \end{aligned}$$

in $H_T^*(F)_S$. Since

$$i^*: H_T^*(M)_S \xrightarrow{\sim} H_T^*(F)_S$$

is an isomorphism, $\gamma = \phi$ in $H_T^*(M)_S$. □

Since $\pi^F: F \rightarrow \{\text{pt}\}$ factors through $\pi^M: M \rightarrow \{\text{pt}\}$,

$$\pi^F = \pi^M \circ i, \quad \pi_*^F = \pi_*^M \circ i_*.$$

(Recall the first case of the Gysin map.)

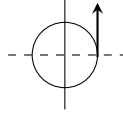
Apply π_*^M to both sides:

$$\boxed{\int_M \phi = \sum_F \int_F \frac{i^* \phi}{e_T(N)}} \quad \text{in } R_S.$$

This is the *Atiyah–Bott* (–*Berline–Vergne*) *localization formula*.

Note that if $\phi \in H_T^*(M)$, the left-hand side belongs to R , hence so does the right-hand side. If $\deg \phi = \dim M$, we get an element of $\mathbb{R}$.

Example. $S^1 \curvearrowright M$, with isolated fixed points. Let $X = -2\pi i u \in \text{Lie}(S^1)$.



$$\phi = \phi_{2n} + \phi_{2n-2}u + \cdots + fu^n \in \Omega(M)^{S^1}[u],$$

where $\dim M = 2n$, and

$$d_X \phi = (d - u \iota_X) \phi = 0.$$

Then

$$\int_M \phi_{2n} = \int_M \phi = \sum_{p \in M^{S^1}} \frac{f(p)}{m_1(p) \cdots m_n(p)},$$

where $m_1(p), \dots, m_n(p)$ are the multiplicities of

$$S^1 \curvearrowright T_p M \simeq L_{m_1} \oplus \cdots \oplus L_{m_n}.$$

Here L_m is the two-dimensional irreducible representation of S^1 :

$$e^{it} \mapsto \begin{pmatrix} \cos mt & -\sin mt \\ \sin mt & \cos mt \end{pmatrix}.$$

(f depends on the choice of u and X .)

Now consider $S^1 \curvearrowright (M, \omega)$ with moment map H .

$$d_X(\omega + uH) = 0 \implies d_X e^{\omega + uH} = 0,$$

where

$$e^{\omega + uH} = e^{uH} \left(1 + \omega + \frac{\omega^2}{2} + \cdots + \frac{\omega^n}{n!} \right).$$

$$\begin{aligned} \int_M e^{uH} \frac{\omega^n}{n!} &= \int_M e^{uH + \omega} \\ &= \sum_{p \in M^{S^1}} \frac{1}{u^n} \frac{e^{uH(p)}}{m_1(p) \cdots m_n(p)}. \end{aligned}$$

The right-hand side is an entire function in u (hence the left-hand side). By analytic continuation, set $u = it$:

$$\int_M e^{itH} \frac{\omega^n}{n!} = \frac{1}{(it)^n} \sum_{p \in M^{S^1}} \frac{e^{itH(p)}}{\prod_{j=1}^n m_j(p)}.$$

Note that

$$dH(p) = 0 \iff \iota_X \omega(p) = 0 \iff X(p) = 0.$$

Thus

$$\text{Crit}(H) = M^{S^1}.$$

Also, around $p \in M^{S^1}$,

$$\omega = \sum_j dx_j \wedge dy_j, \quad X = 2\pi \sum_j m_j(p) (-y_j \partial_{x_j} + x_j \partial_{y_j}).$$

It follows that

$$H = H(p) - \pi \sum_j m_j(p) (x_j^2 + y_j^2), \quad \prod_j m_j(p) \approx \sqrt{\det(\text{Hess}_H(p))}$$

up to sign. □

References: 1. L. Tu. Intro. Lectures on Equiv. Cohom. 2. Mirror symmetry chapter 3, 4.